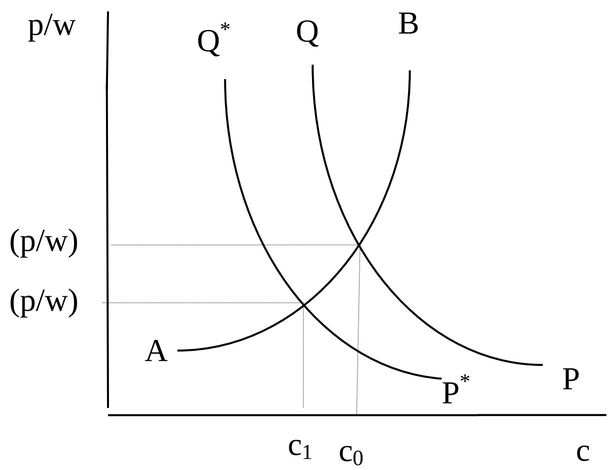

Figure 3.2: Effect of trade on real wage and consumption

The intersection of AB and PQ curves determine the equilibrium values of (p /w) and c. The equilibrium number of products (N) is determined as follows:

$$L = \sum_{i=1}^{N} L_i = \sum_{i=1}^{N} (\alpha + \beta y_i) = N(\alpha + \beta y_i) = N(\alpha + \beta Lc)$$

$$N = \frac{L}{(\alpha + \beta Lc)} = \frac{1}{(\alpha + \beta Lc)/L} = \frac{1}{\left[(\alpha / L) + \beta c\right]} \qquad (6)$$

where L is to total labour force in the country. The assumption of symmetry means that all firms are alike and hence we can simply replace the summation sign $\left(\sum_{i=1}^{N}\right)$ with N. The term y is replaced as L_c as total output of each variety (y) is equal to the total demand for each variety (L_c). Therefore, given L and the fixed parameters α and β, the value of N is endogenously determined once the equilibrium value of c is known. Thus, equilibrium in the closed economy is fully determined.

What will happen when the home country `A' opens trade with the foreign country `B', which is identical with `A' in every way? In the H-O-S setting, there is no reason for these countries to engage in trade as firms in the two countries are symmetric (prices and quantities are identical across varieties in the two countries). But the story is different in the present model in that trade in differentiated commodities will take place even if identical ranges of varieties are produced in both countries in autarky. The reasoning for this goes as follows. In the absence of trade barriers, countries A and B together constitute an `integrated economy' (a term to represent two or more open economies combined), which is equivalent to twice the size of either of the two countries. Firms in this ‘integrated economy’ have no incentive to produce the same varieties as other firms because every variety enters into consumer's utility function symmetrically. Since no two firms produce the same variety, each variety will be produced in only one country. At the same time, all varieties are demanded in both the countries, which necessitate international trade in different varieties of the same commodity Y and hence IIT.

Two identical countries trading is just like doubling the population L, which will have no impact on AB curve. However, the PQ curve will shift down (see equations 4 and 5). As a result of this shift the following changes occur: (i) consumption of each variety falls from c_0 to c_1 ; (ii) p/w falls from $(p/w)_0$ to $(p/w)_1$ which means that real wage w/p rises; (iii) number of varieties (N) available to consumers in each country increases due to the rise in the total number of consumers (L in integrated economy) and fall in the consumption of each variety (see equation 6); (iv) output produced by each firm (y) increases[1]; and (v) number of varieties produced in each country falls. The last result means though the number of varieties available to consumers in each country go up, the number of varieties produced in each country falls. The number of varieties produced should necessarily decline in order to satisfy the employment condition in each country.

$$L = N(\alpha + \beta y)$$

$$N = \frac{L}{(\alpha + \beta y)} \qquad (7)$$

Given that productive resources in each country (L) remains constant, an increase in the value of y means that N should necessarily fall. While the number of varieties produced in each country falls, the sum of varieties from both countries under free trade exceeds the number in any single country before trade. Larger number of varieties for consumption means that consumers in each country clearly gains from trade. Higher real wages are another source of gains from trade in this model. The model shows that trade liberalization leads to a 'scale effect' (surviving firms expand their outputs) as well as a 'selection effect' (some firms are forced to exit).

Ideal variety approach developed by Lancaster (1980) and Helpman (1981) differ from Krugman's model in that the symmetry assumption of the utility function is not used. Instead, in this approach, every consumer residing in the country A is assumed to have preference for an ideal variety. Consumers differ in terms of the most preferred variety, but the assumption of equal density of preferences implies that the same aggregate demand exists for every variety. The supply side is characterised by free entry and exit, identical cost function for all varieties, and the existence of scale economies in the production of every variety. Because of the existence of scale economies, the number of varieties produced cannot be infinite suggesting that some consumers will be unable to get their ideal variety. Firms entering the market must decide on the particular variety and price. Profit maximising behaviour will ensure that no two firms will produce the same

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[1] This is evident from equation (5). Since p/w falls while α and β remain constant , it is clear that Lc = y should necessarily increase.

variety in equilibrium. Also, in equilibrium, each variety will be produced in the same quantity, will be sold at the same price and each firm will earn zero profit. This situation is referred to as perfect monopolistic competition.

Now, consider the possibility of trade with a foreign country B which is identical with `A' in every way. The symmetry assumption will ensure that the autarkic equilibrium features of A and B are identical and the `integrated economy' is equivalent to twice the size of either of the two countries. Also, in the 'integrated economy', each variety will be produced by only one firm and therefore in only one country. All varieties will, however, be consumed in both countries, giving rise to IIT.

The basic models outlined above have been extended by integrating them with the H-O-S approach to international trade. Generally, this has been done by introducing a second output sector (such as, agriculture, producing a homogenous good under conditions of constant returns) and cross-country differences in factor endowments. The integrated models predict that trade based on comparative advantage and trade based on scale economies will take place side by side.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is the effect of intra-industry trade on real wage and consumption?

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2) What will happen when the home country 'A' opens trade with the foreign country `B', which is identical to 'A'?

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3.3.3. IIT in Vertically Differentiated Commodities

The horizontal models are considered to be of greater relevance for understanding the occurrence of bilateral IIT among developed countries. A different approach pertaining to IIT between countries with clearly different factor endowments takes its point of departure in vertical product differentiation. Most economists formulated models of vertical IIT without recourse to economies of scale. In general, these models predict the pattern of IIT along the lines similar to the pattern of inter-industry trade predicted in the conventional trade model, according the central role to factor endowment differences. What follows is a brief discussion of the essential elements of the vertical models.

Falvey (1981) adopted a partial equilibrium approach by concentrating on trade within a single industry, which is assumed to posses a stock of industry specific capital and produces a continuum of products differentiated by quality. Higher quality products are characterised by higher capital labour ratio used in their production. Therefore, the comparative advantages of capital abundant countries lie on the higher ends of the quality spectrum and that of labour abundant countries lie on the lower ends. Vertical IIT between two countries will arise given an overlap in the demand for different qualities and this possibility is enhanced the greater is the factor endowment difference between the countries. Assuming that relative capital abundance is reflected in relative income per capita, the model suggests a hypothesis contrary to what emerges from the horizontal models. That is, the intensity of IIT will be positively correlated with the per capita income differences between the trading partners. Another testable hypothesis, which emerges from the model, is that the shares of IIT vary inversely with the levels of trade restriction of partner countries.

The supply side in the Falvey and Kierzkowski (1987) model is a close descendant of Falvey (1981), the demand side being fully elaborated in the former. Each individual is assumed to demand only one type of differentiated product and given relative prices, this preferred quality is determined uniquely by the individual's income. Since the aggregate income of an economy is not equally distributed, there is, at any point in time, an aggregate demand for a variety of differentiated goods. This holds well in the `integrated economy' as well and consequently vertical IIT will emerge, the intensity of which being higher the more dissimilar (in income distribution) are the trading countries. The model also suggests that the shares of IIT will be positively correlated with the market size of countries.

In the preceding models, capital assumes the decisive role among the factors of production in determining the product specification. Whereas, Flam and Helpman (1987), in their model, ascribe the central role to labour. According to them, population in every country is characterised by a non-degenerate distribution of skills and differences in skills are reflected in differences in the endowment of effective labour supply. Higher quality products are characterised by relatively larger inputs of labour used in their production and the pattern of IIT reflects differences in technology and in income distribution.

A strikingly different mode of analysis focuses on the markets where R&D expenditure (considered a sunk cost) is a prerequisite to quality improvement. Given that the average variable cost rises only slowly with quality, the number of firms in the market is bounded; by extending the distribution of income the upper bound on the number of firms increases, however. Thus, the `integrated economy' can support a large number of firms if the two countries are more dissimilar in income distribution. The higher income country specialises in higher quality products and the lower income country specialises in lower quality products resulting in vertical IIT.

3.3.4. IIT in Intermediate Products

Ethier (1979, 1982) focuses on the significance of three important aspects to international trade: international returns to scale, product differentiation in producer goods and interdependence of world industrial activity. He points out that scale economies resulting from an increased division of labour depend on the size of the world market rather than the production level in any one country. The interdependence of world industrial activity implies global dispersion of the distinct operations consisting of production process in an industry, which in turn depends on the efficiency gains from increased specialisation.

The analysis produces a model of IIT in intermediate products, wherein the desire for product variety arises as a consequence of larger markets permitting greater division of labour. This is in contrast with the models discussed earlier wherein the desire for product variety arises from consumer preference. The analysis further demonstrates that internal scale economies and product differentiation are not the only factors determining the degree of intra-industry trade.

In agreement with the horizontal models, Ethier's model also predicts that similarities of factor endowments between nations tend to promote IIT. Also, if differentiated intermediate goods are tradable, the analysis of trade pattern is similar to that of differentiated final goods.

3.3.5. IIT in Identical Commodities

The usual approach to IIT is to assume that such trade arises because slightly different commodities are produced and traded to satisfy the desire for variety. However, Brander (1981) and Brander and Krugman (1983) have shown that there are reasons to expect IIT even in identical products. In their models, the pattern of trade is determined by the interaction of increasing returns to scale, transport costs, and oligopolistic behaviour of international firms. A Cournot strategy is assumed in which each firm maximises profit assuming the output of other firms in each market (domestic and foreign) remains the same. The market equilibrium involves IIT despite the fact that both countries A and B (symmetric) produce identical commodities, and there is an obvious loss due to transport costs. (Transport costs, however, should not be too high. As transport costs fall, the proportion of foreign goods in domestic consumption increases.) IIT is the result of what is called “reciprocal dumping"(also called "cross-hauling"), i.e., each firm dumps into other firm's home market. The term "dumping" is because the f.o.b price of export is less than the domestic price. The firm's incentive to bear the higher effective marginal cost (including transport costs) associated with exporting stems from the possibility of its marginal revenue in the foreign market exceeding that in the domestic market because the market share of each firm in the foreign country is smaller than that in the home country. This is an equilibrium situation because imperfectly competitive firms set marginal revenue equal to marginal costs in different markets. The drawback of this model is that the Cournot assumption ignores strategic interdependence of firms usually observed in oligopolistic markets. In particular, the ability of this model to explain the pattern of IIT in the context of developing countries is suspect.

Check Your Progress 4

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Intra-Industry trade in Intermediate goods.

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2) What could be the possible reasons behind IIT in identical products?

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3) Discuss the relevance of vertical IIT.

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3.4 MEASUREMENT OF IIT

Two major issues in the empirical analysis of IIT are concerned with: (i) the definition of 'industry' and (ii) measurement of IIT. The measurement issues were subjected to controversies ever since they were first raised and now there exists some professional consensus on most of the issues.

3.4.1 Defining `Industry' and the Problem of Categorical Aggregation

Operationalizing the concept of 'industry' is vital for the empirical analysis of IIT. It is, however, easier said than done, as there is no agreed single definition of 'industry'. Mainly, two criteria have been followed in economic analysis. The first considers different products as belonging to a single industry if they are good substitutes in production, while the second criterion is related to the extent of substitutability of products in consumption.

A major issue for the empirical analysis is that the criteria for official trade classification do not coincide exactly with the criteria for products to constitute IIT. The problem of categorical aggregation arises as a result of the inclusion in a single trade category of products, which either have different production functions or have different end uses. Importantly, the extent of IIT, as estimated by the usual measures is sensitive to the level of aggregation (of products) which defines an industry: higher levels of aggregation usually give rise to greater extent of recorded IIT. The use of a more detailed classification system for the measurement of IIT does not necessarily solve the problem because it would tend to separate commodities that are good substitutes in production and consumption.

Thus, the degree and kind of homogeneity of commodity items under each trade category need to be considered for the purpose of empirical analysis. The most common practice is to select a specific level of statistical aggregation in the official classification as the best approximation to the concept of industry. Now, there exists a professional consensus to consider the products grouped under 4 or 6-digit level of Harmonized System (HS) of Classification as roughly corresponding with the concept of an industry. In general, commodities grouped under a particular 4 or 6-digit code may be close substitutes in production, consumption or both. First, there may be products traded having similar input requirements, are close substitutes in production, but not in consumption. Secondly, there are products traded that are close substitutes in consumption, but not in production. A third category involves very similar goods, which are close substitutes in consumption and production. Finally, the exchange of absolutely identical products can also be found. In practice, the simultaneous exchange in these four types of commodities is considered as IIT, while other types of transactions constitute inter industry trade.

3.4.2 Measuring the Intensity of IIT

The intensity of IIT is usually measured by using the well-known Grubel Lloyd (GL) index:

$$GL_i = \frac{(X_i + M_i) - |X_i - M_i|}{(X_i + M_i)} \times 100, \qquad (8)$$

where GL_i is the IIT index for industry i, and X_i and M_i are values of exports and imports in industry i. The value of GL_i ranges from 0 to 100. If there is no IIT (i.e., one of X_i or M_i is zero) GL_i takes a value 0. If all trade is IIT (i.e., $X_i = M_i$), GL_i takes a value of 100. Grubel and Lloyd (1975) proposed the following weighted index to arrive at an aggregate measure of IIT:

$$GL = \frac{\sum[(X_i + M_i) - |X_i - M_i|]}{\sum(X_i + M_i)} \times 100. \qquad (9)$$

The following features of Grubel Lloyd index are worth noting.

1) GL_i is independent of the absolute value of exports and imports, thus it does not allow conclusion on the absolute value of IIT.

2) GL_i is symmetric in the rise in exports and imports, thus it could rise due to a rise in either exports or imports or both.

3) GL_i is non-linear in changes, so a small alteration in the coefficient would disguise a big change in trade volumes and vice versa.

4) GL_i is subject to aggregation problems. Grubel and Lloyd argue that "…the measure of IIT at a more aggregative level is greater than, or at least no less than, the measured intra-industry trade with a finer commodity breakdown".

5) Grubel and Lloyd observed that GL is a biased downward measure of IIT if the country's total commodity trade is not balanced or if the mean is an average of some subset of all industries for which exports are not equal to imports. They considered this an undesirable feature of a measure of average IIT and suggested the need for correcting for trade imbalance. They proposed to use an adjusted measure. Although, many economists disagree with the idea of adopting adjusted measure.

Check Your Progress 5

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Grubel Lloyd (GL) index for Measuring the Intensity of IIT.

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2) What were the major issues in the empirical analysis of IIT?

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3.5 LET US SUM UP

In this unit, we discussed about Intra industry trade and how trade liberalization has supported its growth. IIT means the simultaneous export and import of products, which are very close substitutes for each other in terms of factor inputs and consumption. Growth of IIT is an outcome of greater specialisation in the manufacturing of unique varieties or product lines by the individual plants in different countries. In a liberalized environment, intensity of IIT has increased in most of the countries. In absence of trade barriers, IIT has increased in many industries.

We also distinguished between the two types of IIT – horizontal and vertical IIT. Horizontal IIT refers to the simultaneous export and import of the differentiated products within an industry which are of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities. While discussing the theories of Intra-Industry Trade, we also discussed about the need for separate theory for Intra-Industry Trade. We discussed about 'love for variety' and 'ideal variety approach' in case of horizontal IIT. We also discussed about IIT in identical goods and intermediate goods. We further discussed about the major issues in the empirical analysis of IIT and Grubel Lloyd (GL) index for measuring the Intensity of IIT.

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